WORKED SOLUTION

STEP III 2001, Question 12

Probability and Statistics • Expectation by Indicators

STEP III 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q12

The key idea

In part (i), don't try to find the distribution of the number of black balls: break the count into one indicator per black ball and use linearity of expectation. Drawing without replacement until the bag is empty is just a random ordering of all the balls, so each indicator comes down to a symmetry argument. In part (ii) the bag never changes between white draws, so the wait for each white ball is geometric, and the "hence" comes from noticing that after a white ball is removed we are back in the same situation with one fewer white ball.

Solution

Part (i)

Imagine continuing to draw until the bag is empty. Since every draw is at random and nothing is returned, the order in which the $b+w$ balls come out is a uniformly random ordering of them. The black balls drawn by the time the last white ball is removed are exactly the black balls that come before the last white ball in this ordering.

Label the black balls $1, 2, \ldots, b$ and let $I_j = 1$ if black ball j comes out before the last white ball, and $I_j = 0$ otherwise. If N is the number of black balls drawn when the last white ball is removed, then

$$N = I_1 + I_2 + \cdots + I_b, \qquad \text{so} \qquad \mathrm{E}(N) = \sum_{j=1}^{b} \mathrm{P}(I_j = 1)$$

by linearity of expectation (which needs no independence between the I_j).

? **Ask yourself** What is the chance that black ball j comes after every white ball?

Look only at the $w+1$ balls consisting of black ball j and the w white balls. In a uniformly random ordering of all the balls, the relative order of these $w+1$ is also

uniformly random, so black ball j is equally likely to be in any of the $w+1$ positions among them. It is last of the group with probability $\frac{1}{w+1}$, and that is precisely the event $I_j = 0$. Hence

$$\mathrm{P}(I_j = 1) = 1 - \frac{1}{w+1} = \frac{w}{w+1},$$

and

$$\boxed{\mathrm{E}(N) = \frac{bw}{w+1}.}$$

Part (ii)

Now black balls go back, so until a white ball is drawn the bag always holds b black and w white balls. Each draw is therefore black with probability $p = \frac{b}{b+w}$ and white with probability $q = \frac{w}{b+w}$, independently of earlier draws, since the contents are the same each time.

Let X be the number of black draws before the first white ball. Then $X = n$ means n blacks followed by a white, so

$$\mathrm{P}(X = n) = p^n q, \qquad n = 0, 1, 2, \ldots$$

If $b = 0$ then $X = 0$ and the result is trivial, so take $b \geq 1$; then $0 < p < 1$, and using $\sum_{n\geq 1} n x^{n-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$ (differentiate the geometric series),

$$\mathrm{E}(X) = \sum_{n=0}^{\infty} n p^n q = pq \sum_{n=1}^{\infty} n p^{n-1} = \frac{pq}{(1-p)^2} = \frac{p}{q} = \frac{b/(b+w)}{w/(b+w)} = \frac{b}{w},$$

since $1 - p = q$. This is the required result.

For the last white ball, write the total number of black draws as $X_w + X_{w-1} + \cdots + X_1$, where X_k is the number of black draws made while exactly k white balls are in the bag. While k white balls remain, the bag holds b black and k white balls, so this stage is exactly the first-white problem with w replaced by k, and the result just proved gives $\mathrm{E}(X_k) = \frac{b}{k}$. By linearity of expectation the expected number of black draws when the last white ball is removed is

$$\boxed{\sum_{k=1}^{w} \frac{b}{k} = b\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{w}\right).}$$

Checks

With $w = 1$ both parts are easy to see directly. In (i), the single black ball lands before the white one with probability $\frac{1}{2}$, giving $\frac{b}{2}$, which matches $\frac{bw}{w+1}$. In (ii), there is one stage only and the sum is just $\frac{b}{1} = b$. As $w \to \infty$ in (i), almost every black ball comes out before the last white one, and indeed $\frac{bw}{w+1} \to b$.

We checked (i) by listing every ordering for all $b \leq 4$, $w \leq 4$; for $b = 3$, $w = 4$ it gives exactly $\frac{12}{5}$. For (ii), a simulation of 2×10^5 runs with $b = 3$, $w = 4$ gave a mean of about 6.245, against $3(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) = \frac{25}{4} = 6.25$.

Common slips

- In (i), assuming the black balls drawn are independent of one another. Linearity of expectation doesn't need independence, which is why indicators work so well.
- In (ii), using the geometric distribution that counts the white draw too, which gives $\frac{1}{q} = \frac{b+w}{w}$, one more than the answer.
- Using $b + w$ in the denominator at every stage of the last part: the number of white balls in the bag drops by one at each stage.

Beyond STEP

The second part is the coupon collector's problem in disguise: summing geometric waiting times over stages gives a harmonic sum, which grows like $\ln w$. Writing a count as a sum of indicators, as in part (i), is one of the most powerful techniques in probability, and you will meet it again when computing expected numbers of fixed points, records and cycles in random permutations. Blitzstein and Hwang, *Introduction to Probability*, has many examples of both ideas.

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